Reg. No: ……

Name: ……

**MAHATMA GANDHI UNIVERSITY, KOTTAYAM**

**MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2026**

**FOURTH SEMESTER**

**DSE- MG4DSEBIF201- CELLULAR ENZYMOLOGY**

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

**Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest() and Appreciation(Ap)**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

**Part A**

Short Answer Questions

Answer any 5 out of 7 Questions

Each Question carries 2 marks

1. Explain transferases with one example [U]/[CO1]
2. Describe Reversible enzyme inhibition [U]/[CO3]
3. What is meant by an apoenzyme? [K]/[CO2]
4. Name any two coenzymes involved in oxidation-reduction reactions [U]/[C04]
5. State the significance of enzymes on reducing activation energy [U]/[C05]
6. Define diagnostic enzymes. [K]/[C06]
7. Differentiate between apoptosis and necrosis based on their biological consequences. [An]/[C07]

[5x2=10]

**Part B**

Short Essay Type Questions

Answer any 6 out of 8 Questions

Each Question carries 5 marks.

8. Define enzyme as biological catalysts with suitable examples [K]/[CO2]
9. Describe ribozymes and its biological significance. [U]/[CO4]
10. Explain the effect of temperature on enzyme activity [U]/[C04]
11. Explain the Michaelis- Menton model of enzyme action [U]/[CO5]
12. Describe the role of enzymes in drug discovery and development. [U]/[CO6]
13. Describe enzyme turn over number. [U]/[CO3]
14. Analyze the working of the $Na^{+}$–$K^{+}$ pump and explain why it is

essential for cellular function.
[An]/[C07]

15. Explain the lock and key model of enzyme–substrate interaction with a diagram. [U]/[CO1]

(6x5 =30)

**Part C**

Long Essay Type Questions
Answer any 3 out of 5 Questions
Each Question carries 10 marks

16.Explain the Lineweaver-Burk plot, its derivation and importance in enzyme kinetics. [U]/[CO5]

17. Explain the effect of inhibitors and activators on enzyme. activity, classify inhibitors and describe their mechanism of action. [U]/[CO4]

18. Recall the concept of personalized medicine with special reference to enzyme polymorphism and pharmacogenomics. [K]/[CO6]

19.Compare and contrast structural organization of prokaryotic and eukaryotic cells. [An/CO7]

20.Summarize enzyme inhibition and it's types. [U]/[CO3]

(3x10 = 30)